Roop Yaswanth Nagabhairava

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RESEARCH EXPERIENCE

University of Florida, Smart Autonomy Lab Gainesville, FL
Research Engineer Aug 2026 - Present

- **Binary code similarity:** Co-developing a novel method to quantify whether a patched or recompiled binary preserves the original's functional and security behavior without source code; own baselines, model training, and evaluation across compiler, optimization-level, and architecture variants.
- **Cross-embodiment manipulation:** Developing methods to transfer manipulation policies learned on one gripper to collaborative robot arms with different gripper geometries; testing whether trajectory resampling adapts demonstrations to new grippers.
- **Hierarchical VLM-to-VLA control:** Designing a framework where a vision-language model turns complex instructions into simple sub-task prompts for a Vision-Language-Action policy, with a verifier for failure recovery.

RESEARCH WRITING

R. Y. Nagabhairava, "CineMatch: A Deployed Multilingual Hybrid Recommender with Leakage-Resistant Temporal Evaluation," manuscript, 2026.

PROJECTS

CineMatch | Hybrid Recommender with Graph Contrastive Learning roop-yaswanth.github.io/work/cinematch
Python, PyTorch, RecBole, BGE-M3, FAISS, SciPy

- Fused multilingual BGE-M3 semantic retrieval (1.37M titles) with XSimGCL graph-contrastive collaborative filtering (MovieLens-32M). Designed a leakage-resistant temporal evaluation with frozen catalogs, SHA-256 artifact fingerprints, and Holm-corrected paired tests.
- On 2,500 held-out users (19,419 future positives), reached 0.066 NDCG@10, about twice MostPopular and LightGCN (Holm-corrected $p < 1e\text{-}29$); DPP raised cross-language exposure from 0.13 to 0.30 for 1,013 viewers with NDCG@10 unchanged. Built a Next.js/FastAPI product with rating-driven discovery and language controls.

SoundVector | Contrastive Embeddings for Music Retrieval roop-yaswanth.github.io/work/soundvector
Python, PyTorch, hnswlib, scikit-learn, Gemini API

- Trained 256-d contrastive track embeddings for 899K songs and built HNSW retrieval (99.95% recall@10 vs exact search) with a ranker for artist variety and recording deduplication; on 1,000 random seeds it raised distinct artists per top 10 from 2.8 to 9.0 and cut lists repeating a recording from 35% to 0, keeping 89% of picks in the seed's genre.
- Built feedback-driven personalization in which active saves and completed previews shape the queue, removing a like rebuilds the profile, and dislikes filter recordings across catalog and provider IDs.

Multi-Image Super-Resolution for Prostate MRI roop-yaswanth.github.io/work/mri
Python, PyTorch, MONAI, torchmetrics, pydicom, NumPy, scikit-image

- Built a slice super-resolution pipeline on 782 patients (TCIA Prostate-MRI-US-Biopsy); SwinIR reached 28.18 dB PSNR and 0.925 SSIM on 8,587 held-out slices, with the highest mean scores among the three evaluated configurations.

EDUCATION

University of Florida, Gainesville, FL May 2026
Master of Science in Applied Data Science | GPA: 3.90/4.00
Coursework: Applied Machine Learning I & II, Deep Learning in Medical Image Analysis, Math Foundations for Data Science I & II

K L University, Vijayawada, India May 2024
Bachelor of Technology in Artificial Intelligence and Data Science | GPA: 9.28/10.00

SKILLS

Research Areas: binary code similarity, program analysis, robot manipulation, multimodal vision-language and vision-language-action models, graph and contrastive representation learning, recommender systems
Machine Learning: PyTorch, Hugging Face Transformers, RecBole, scikit-learn, TensorFlow/Keras, MONAI, LoRA/QLoRA fine-tuning, FAISS, HNSW
Engineering: Python, SQL, TypeScript, Linux, FastAPI, PostgreSQL, MongoDB, NumPy, Pandas, SciPy
Cloud & Tools: AWS, Azure, Google Cloud, Docker, Git, GitHub Actions, Claude Code (agentic coding)
Certifications: Google Cloud Associate Cloud Engineer; Microsoft Azure AI Fundamentals (AI-900)